

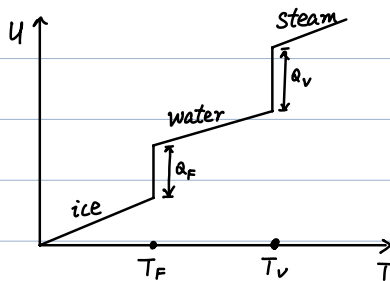
Heat

Heat capacity (recall last class):

$$C = \frac{\Delta Q}{\Delta T}$$

$$\Delta Q \text{ absorbed} \Rightarrow \Delta T = \frac{\Delta Q}{C} \text{ increased}$$

Exception: Phase transitions of H_2O



$$\text{At } \begin{cases} T_F \text{ (Freezing point)} \\ T_V \text{ (Vaporizing point)} \end{cases}, \quad \begin{cases} Q_F \\ Q_V \end{cases} \text{ absorbed, but } T \text{ unchanged!}$$

Latent heat (相变潜热): Q_F, Q_V

$$Q = L M$$

$\left\{ \begin{array}{l} L: \text{heat of transformation} \\ M: \text{mass} \end{array} \right.$

$$Q_F = L_F M, \quad L_F = 333 \text{ kJ/kg} \quad \text{heat of fusion (溶解热)}$$

$$Q_V = L_V M, \quad L_V = 2256 \text{ kJ/kg} \quad \text{heat of vaporization (汽化热)}$$

Unit for heat

specific heat of water:

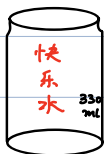
$$4184 \text{ J} \cdot \text{kg}^{-1} \cdot \text{K}^{-1} = 4.184 \text{ J} \cdot \text{g}^{-1} \cdot \text{K}^{-1}$$

Small calorie (gram-calorie):

$$1 \text{ cal} \equiv 4.184 \text{ J}$$

Grand calorie (kilogram-calorie):

$$1 \text{ Cal} \equiv 4184 \text{ J}$$

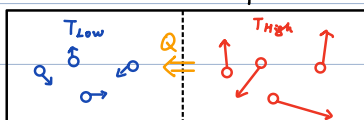


Each can: 140 Cal $\rightarrow$ raise 1 kg of water by 140 K

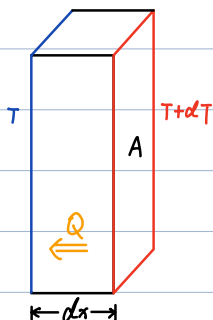
Heat: Energy transfer across boundary.

$\downarrow$
Molecular kinetic energy

Heat conduction (microscopic)



Heat conduction (macroscopic)



Fourier heat conduction law:

$$j_q \equiv \frac{dQ}{A dt} = -k \frac{dT}{dx}$$

$\downarrow$ heat current $\downarrow$ thermal conductivity

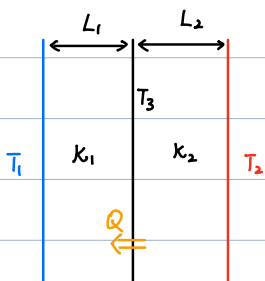
A: area

"-": direction of j_q opposite to $\frac{dT}{dx}$

"k": material-specific

e.g. $k_{\text{metal}} \gg k_{\text{wood}}$

Example



Q: T_3 ? j_q ?

$$j_q = -k_1 \frac{T_3 - T_1}{L_1} = -k_2 \frac{T_2 - T_3}{L_2}$$

$$\Rightarrow T_3 = \frac{k_1 T_1 / L_1 + k_2 T_2 / L_2}{k_1 / L_1 + k_2 / L_2}$$

$$\Rightarrow j_q = - \frac{T_2 - T_1}{L_1 / k_1 + L_2 / k_2}$$

Mean free path

Average distance between collisions.

$$l = \frac{1}{\sqrt{2} \pi d^2 N/V}$$

d : diameter of molecules

N/V : # of molecules in unit volume

Proof:

relative speed
↑

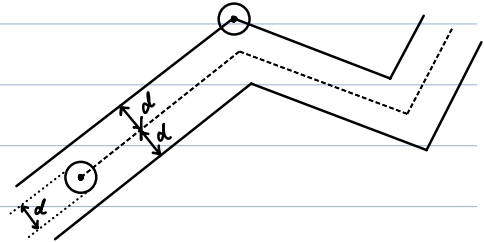
Volume of tube: $\pi d^2 \cdot \bar{v}_r t$

$\Rightarrow$

of molecules in tube: $\frac{N}{V} \cdot \pi d^2 \bar{v}_r t$
(# of collisions)

$\Rightarrow$

Mean free path:
$$l = \frac{\bar{v} t}{\frac{N}{V} \pi d^2 \bar{v}_r t}$$
$$= \frac{\bar{v}/\bar{v}_r}{\pi d^2 N/V}$$



How to get $\bar{v}_r$?

Effective mass:

$$m' = \frac{m_1 m_2}{m_1 + m_2} = \frac{m}{2}$$

Maxwell speed distribution:

$$\begin{cases} f(v) = 4\pi \left(\frac{m}{2\pi k_B T} \right)^{3/2} e^{-\frac{m v^2}{2 k_B T}} v^2 \\ f(v_r) = 4\pi \left(\frac{m'}{2\pi k_B T} \right)^{3/2} e^{-\frac{m' v_r^2}{2 k_B T}} v_r^2 \\ = 4\pi \left(\frac{m}{4\pi k_B T} \right)^{3/2} e^{-\frac{m v_r^2}{4 k_B T}} v_r^2 \end{cases}$$

$\Rightarrow$

$$\begin{aligned} \bar{v}_r &= \int_0^\infty v_r f(v_r) dv_r \\ &= \int_0^\infty v_r \cdot 4\pi \left(\frac{m}{4\pi k_B T} \right)^{3/2} e^{-\frac{m v_r^2}{4 k_B T}} v_r^2 dv_r \\ &\stackrel{v_r = \sqrt{2} \tilde{v}}{=} 4 \cdot \frac{2^{3/2}}{4^{3/2}} \int_0^\infty 4\pi \left(\frac{m}{2\pi k_B T} \right)^{3/2} e^{-\frac{m \tilde{v}^2}{2 k_B T}} \tilde{v}^3 d\tilde{v} \\ &= \sqrt{2} \bar{v} \end{aligned}$$

$$\Rightarrow l = \frac{1}{\sqrt{2} \pi d^2 N/V} \quad \text{Proved}$$

$$pV = N k_B T$$

$\Rightarrow$

$$l = \frac{k_B T}{\sqrt{2} \pi d^2 p}$$

③

Example. Air at 20°C

$$\frac{N}{V} = \frac{P}{k_B T} = \frac{1.01 \times 10^5 \text{ N/m}^2}{1.38 \times 10^{-23} \text{ (J/K)} \times 293 \text{ K}}$$

$$= 2.50 \times 10^{25} \text{ m}^{-3}$$

Estimate $d = 2 \times 10^{-10} \text{ m}$

$$\Rightarrow l = \frac{1}{\sqrt{2} \pi d^2 N/V}$$

$$= \frac{1}{\sqrt{2} \pi \times 4 \times 10^{-20} \times 2.5 \times 10^{25}}$$

$$= 2.25 \times 10^{-7} \text{ m}$$

Compare to average molecular separation:

$$\sqrt[3]{\frac{1}{N/V}} = 3.4 \times 10^{-9} \text{ m} \quad (< l)$$

Collision frequency:

$$f = \frac{\bar{v}}{l} = \frac{\sqrt{\frac{8k_B T}{\pi m}}}{l}$$

$$= \frac{471 \text{ m/s}}{2.25 \times 10^{-7} \text{ m}}$$

$$\approx 2 \times 10^9 \text{ s}^{-1}$$

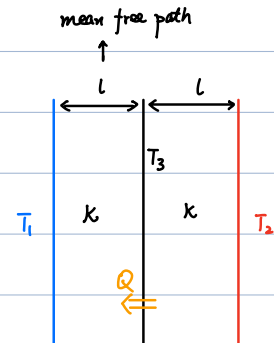
$$\begin{aligned} & N_2 \\ & \uparrow \\ & 28.014 \text{ g/mol} \\ m &= \frac{28.014 \text{ g/mol}}{6.02 \times 10^{23} \text{ mol}^{-1}} \\ &= 4.65 \times 10^{-26} \text{ kg} \end{aligned}$$

(*) Thermal conductivity again:

$$j_Q = \frac{dQ}{A dt} = -k \frac{dT}{dx}$$

$$\begin{cases} \frac{dT}{dx} = \frac{T_2 - T_1}{2L} \\ \frac{dQ}{A dt} = N_{1 \rightarrow 2} \cdot \frac{f}{2} k_B T_1 - N_{1 \leftarrow 2} \cdot \frac{f}{2} k_B T_2 \end{cases}$$

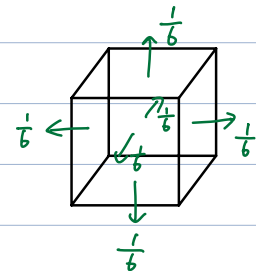
$$f = t + r + 2s \quad (\text{DOF})$$



$N_{1 \rightarrow 2}$: # of molecules from 1 to 2 in unit time & unit area

$$\approx \frac{1}{6} \left(\frac{N}{V} \right)_1 \cdot (A \bar{v}_1 dt)$$

$$= \frac{1}{6} \left(\frac{N}{V} \right)_1 \bar{v}_1$$



$$\Rightarrow \frac{dQ}{A dt} \approx \frac{1}{6} \left(\frac{N}{V} \right)_1 \bar{v}_1 \cdot \frac{f}{2} k_B T_1 - \frac{1}{6} \left(\frac{N}{V} \right)_2 \bar{v}_2 \cdot \frac{f}{2} k_B T_2$$

$$\approx \frac{1}{6} \frac{N}{V} \bar{v} \cdot \frac{f}{2} k_B (T_1 - T_2)$$

$$= \frac{1}{6} \frac{1}{V} \bar{v} C_V (T_1 - T_2)$$

$$= \frac{1}{6} P \bar{v} C_V (T_1 - T_2)$$

$$\left(\frac{N}{V} \right)_1 \bar{v}_1 \approx \left(\frac{N}{V} \right)_2 \bar{v}_2 \approx \frac{N}{V} \bar{v}$$

$$C_V = \frac{f}{2} N k_B$$

$$C_V = \frac{C_V}{M}, \quad P = \frac{M}{V}$$

(4)

In total :

$$\frac{1}{6} \rho \bar{v} c_v (T_1 - T_2) = -\kappa \frac{T_2 - T_1}{2L}$$

$$\Rightarrow \boxed{\kappa = \frac{1}{3} \rho \bar{v} L c_v}$$